Predictability of European Blocking and Storm Daniel Linked to Hurricane Franklin's Extratropical Transition

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Introduction

This study examines the predictability of a chain of events leading to the formation of a devastating Mediterranean cyclone, Storm Daniel. Hurricane Franklin recurved into the midlatitudes and underwent extratropical transition (ET) on 1 September 2023 (Figure 1a). A ridge downstream of Franklin (Figure 1b) developed into an Omega block over Europe on 5 September 2023, while a PV streamer became cut off over the eastern Mediterranean (Figure 1c), leading to the formation of Daniel. Using ECMWF and Met Office ensemble forecasts from the TIGGE archive, together with ERA5 reanalysis we examine the impact of Franklin's ET and associated diabatic processes on downstream high impact weather. We apply a block error metric to relate the predictability of European blocking to Franklin and assess how block accuracy influenced the subsequent development of Storm Daniel.

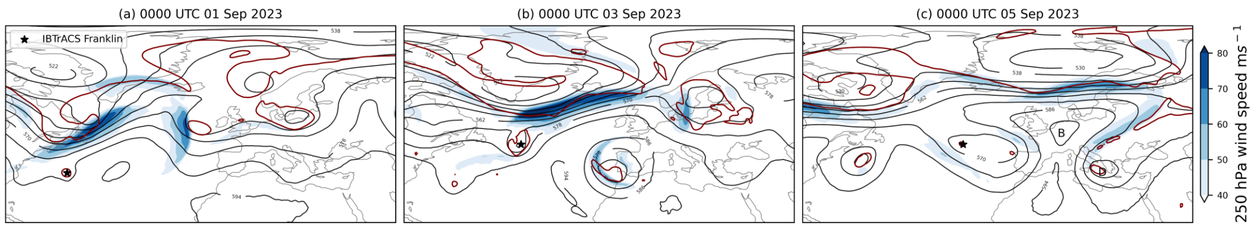


Figure 1: The synoptic evolution of the case. Z500 (grey contours) 250hPa winds (blue shading) and 2PVU on 320K (red contour) at 0000 UTC 1 Sep, 2 Sep and 5 Sep.

Data and Methods

Blocking over Europe is detected using an anomaly-based block detection method, with blocks characterised as a large, positive Z500 anomaly ($Z'_* > 100m$) from the zonal mean. Monthly mean climatology is removed, accounting for seasonal Rossby wave patterns (Suitters et al, 2023).

$$Z'_*(\lambda, \phi, t) = Z_*(\lambda, \phi, t) - \overline{Z}_*(\lambda, \phi, t) \qquad (1)$$

Block error is quantified using the *ZALO* metric, like *PAL* derived by (Wernli et al, 2008) for forecast verification of precipitation. The indices *Z*, *A*, *L* and *O* measure error in block intensity, area, location and overlap of the forecast block compared to reanalysis, respectively. These metrics are combined, accounting for error in all block characteristics:

$$ZALO = \sqrt{Z^2 + A^2 + L^2 + O^2} \qquad (2)$$

ZALO is then used as a response function (*J*) in ensemble sensitivity analysis, to attribute block error to upstream features. Sensitivity (*S*) at each grid point is calculated as,

$$S = m \times \sigma_x \times \alpha, \qquad (3)$$

where m is the slope of the least squares fit of *J* on the precursor field, *x,* and σ_x is the standard deviation of *x*. α acts as a correction factor filtering out sensitivity with weak correlation. In this case, sensitivity is damped when $r^2 < 0.15$, as used in similar midlatitude sensitivity studies (Maddison et al, 2019).

Results

Figure 2 shows how block error (*ZALO*) converges with forecast lead time to block onset (0000 UTC 5 September) for ECMWF and UKMO ensembles. Block error decreases rapidly between lead times of 144 and 120 h (forecasts initialised at 0000 UTC 30 and 31 August). The forecasts initialised on 31 Aug will be used to explain poor predictability.

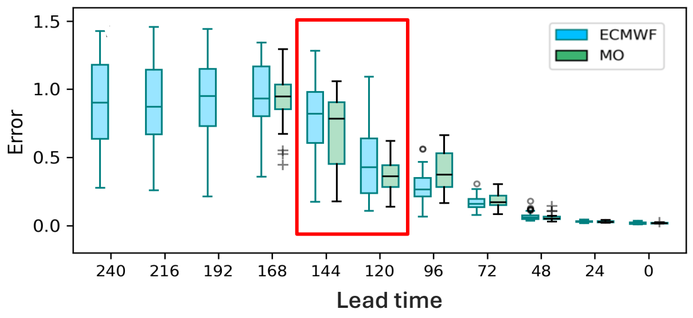


Figure 2: Summarising predictability – box-and-whiskers of *ZALO* with decreasing lead time to block onset (0000 UTC 5 Sep 2023) for ECMWF and UKMO ensemble forecasts. *ZALO* of 0 indicates a block equivalent to ERA5.

Figure 3 shows the ECMWF and UKMO ensemble positions of Franklin, with reference to IBTrACS for these two forecasts at lead times of 48 and 72 h. In the ECMWF, a slow propagation bias is evident at the time of, and after, ET. In the UKMO, three recurvature scenarios are evident: a stall of Franklin, Franklin following IBTrACS, or Franklin recurving northwards too quickly. The members that produce the best blocking place Franklin near IBTrACS, and those that produce the worst blocks stall and don't interact with the midlatitude flow or recurve too quickly.

Figure 4 shows ensemble sensitivity to block error (*ZALO*) on 2, 3 and 5 September for forecasts initialised on 30 and 31 August. Strong sensitivity is observed at short lead times in both forecasts, indicative of block error being highly sensitive to small errors in Franklins position, consistent with Figure 3.

Franklin, in good members, is observed to excite a Rossby wave packet, indicated by the alternating sign of sensitivity, an amplified ridge immediately downstream of Franklin and troughing north of Scandinavia (Figure 4c, d). On 5 September, in the 31 August forecast, the best members produce amplified blocking with a cut-off low in the correct position (Figure 4f).

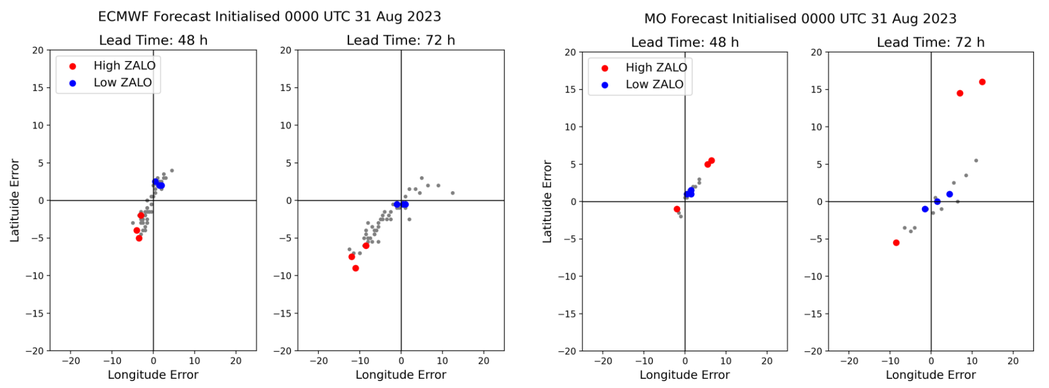


Figure 3: Franklin's position in ECMWF (left) and UKMO (right) ensemble forecasts initialised on 31 Aug at 48 and 72 h lead times. Members with the highest and lowest ZALO are highlighted.

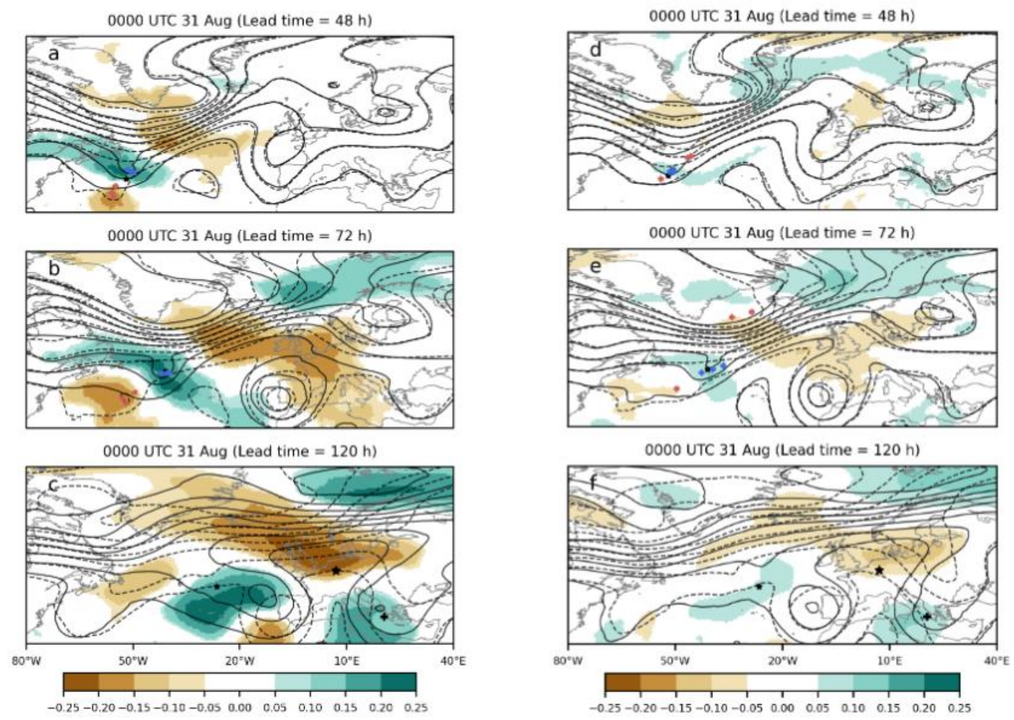


Figure 4: Z500 sensitivity to ZALO for an ECMWF (left) and UKMO (right) forecast on 2 Sep (top), 3 Sep (middle) and 5 Sep (bottom). Contours of the Z500 mean of the three members with the highest (dashed) and lowest (solid) are shown.

The sensitivity signal, evident in the UKMO forecast at a lead time of 48 h (Figure 4d), highlights a short-wave trough ejected from Canada as a region of negative sensitivity. This suggests low block error is produced by members with a weaker trough in this region and that Franklin's positional errors at short lead times, and subsequent downstream block accuracy, may have been influenced by this feature.

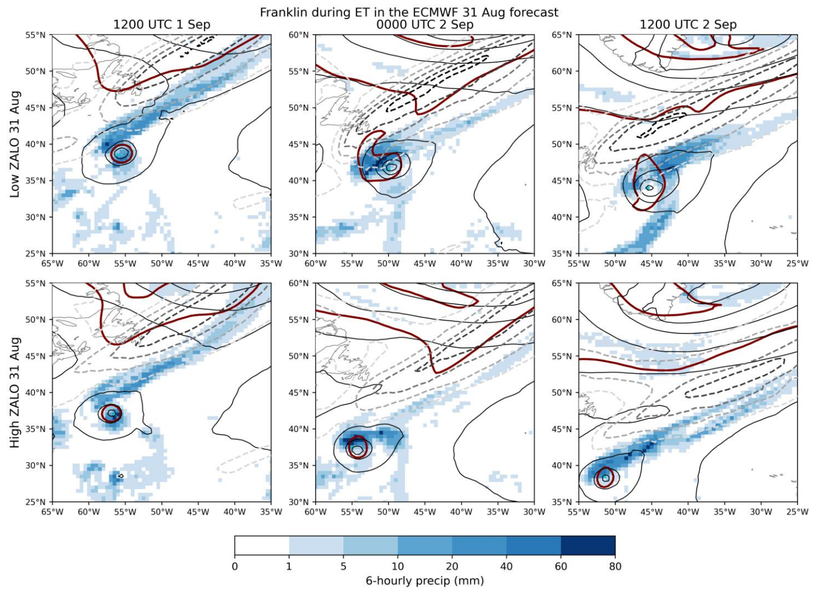


Figure 5: Mean 6-hourly total precipitation (blue shading), jet level (200-300 hPa) winds (dashed contours), and the 2PVU contour ensemble member with the highest and lowest ZALO for the ECMWF forecast initialised 31 Aug at 12-hourly time steps.

In the best ensemble members, Franklin intensifies during transition (Figure 5). These are the members that also exhibit enhanced convective precipitation, anchor to the downstream ridge and produce accurate blocking. The worst members, positioned too far south from the jet, do not reintensify nor maintain downstream ridge amplification.

Conclusions

Ensemble sensitivity analysis identifies Franklin's location as a leading source of downstream forecast uncertainty. Accurate prediction of Franklin's track was critical for developing upper-level ridging and European blocking. In ensemble members that best represented the European block, Franklin remained anchored to a downstream ridge, excited a Rossby wave packet over Europe, and initiated a wave breaking event that produced the cut-off leading to Storm Daniel's development.

Only in those members where Franklin's position was well represented did the storm reintensify, producing enhanced convection and a more accurate representation of downstream blocking. The strength of Franklin's midlatitude interaction during ET is therefore highly sensitive to small track errors, which may be linked to Franklin's interaction with a short-wave trough.

In our future work, we aim to isolate the effects of Franklin's position and latent heating. We are conducting model experiments designed to explore the links between diabatic processes, block predictability, and the representation of downstream high-impact weather.

References

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